Mamba: Selective State-Space Models From Mathematical Foundations to Linear-Time Implementation

Hyperparameter Analysis & Cost Benchmarking on NVIDIA A100

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Abstract

Transformers dominate modern sequence modeling but suffer from $O(n^2)$ complexity in self-attention. Mamba introduces a selective state-space model (SSM) that replaces fixed linear time-invariant (LTI) dynamics with input-dependent parameters. Ensuring content-aware reasoning in $O(n)$ time. This paper presents the mathematical foundations of Mamba, details the selective scan mechanism and hardware-aware implementation, and provides empirical hyperparameter comparisons between two model configurations. One configuration with fully dynamic B and C matrices and another with fixed B and C. The NVIDIA A100-80GB GPU was used to run all experiments. We evaluate on a character-level language modeling task, Sequential CIFAR-10, ListOps, Selective Copying, reporting wall-clock training time and task-specific metrics.

Keywords: Selective state-space models, Mamba architecture, Linear time-invariant systems, Sequence modeling complexity, Character-level language modeling

1 Introduction

The Transformer architecture [Vaswani et al., 2017] has become the dominant paradigm across natural language processing, computer vision, and audio modeling. Its core mechanism, multi-head self-attention, computes pairwise interactions among all tokens in a sequence, yielding $O(n^2)$ time and memory complexity with respect to sequence length n. This quadratic cost manifests in three practical bottlenecks: (1) an inference-time KV-cache that grows linearly per layer per token, (2) memory consumption that prohibits long-context windows on commodity hardware, and (3) throughput degradation as sequences scale beyond a few thousand tokens.

Structured state-space models (SSMs) offer an alternative by modeling sequences through latent state dynamics derived from continuous-time linear systems. Prior SSMs such as S4 and S5 achieve $O(n)$ or $O(n \log n)$ complexity via global convolution, but their *linear time-invariant* (LTI) parameterization treats every input identically, limiting expressiveness on content-dependent tasks.

Mamba [Gu and Dao, 2023] resolves this tension by introducing a **selective scan**: the discretization step size Δ_k, the input projection B_k, and the output projection C_k are all functions of the current input x_k. A hardware-aware parallel associative scan replaces the naïve sequential recurrence,

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preserving GPU efficiency. The result is a model that matches or exceeds Transformer quality at a fraction of the compute.

2 Continuous-Time SSM Foundations

State-space models originate in control theory as linear time-invariant (LTI) systems. The continuous-time formulation is:

$$\dot{h}(t) = A\,h(t) + B\,x(t), \tag{1}$$

$$y(t) = C\,h(t) + D\,x(t), \tag{2}$$

where $h(t) \in \mathbb{R}^N$ is the latent state, $x(t) \in \mathbb{R}$ is the scalar input, $y(t) \in \mathbb{R}$ is the output, and (A, B, C, D) are parameter matrices of appropriate dimension. In practice, D is absorbed into a skip connection, so we focus on $A \in \mathbb{R}^{N\times N}$, $B \in \mathbb{R}^{N\times 1}$, and $C \in \mathbb{R}^{1\times N}$.

The state $h(t)$ acts as a compressed summary of the input history. The transition matrix A governs how the state evolves over time, while B controls how new inputs are incorporated.

2.1 LTI Applications

LTI systems appear across engineering: flight-control autopilots, robotic manipulator dynamics, signal processing filter design, and circuit analysis all rely on the same (A, B, C, D) framework. The mathematical maturity of this formulation provides stability guarantees and well-understood spectral properties that carry over into the neural network setting.

3 Discretization and Dual Computation Modes

Neural networks operate on discrete token sequences. We discretize Equations (1)–(2) via the *zero-order hold* (ZOH) with learnable time-step $\Delta > 0$:

$$\bar{A} = e^{A\Delta}, \tag{3}$$

$$\bar{B} = (A\Delta)^{-1}(e^{A\Delta} - I)\,B. \tag{4}$$

The discretized system admits two equivalent computational views.

Discrete Recurrence (RNN mode). For autoregressive inference, the state update is:

$$h_k = \bar{A}\,h_{k-1} + \bar{B}\,x_k, \quad y_k = C\,h_k. \tag{5}$$

This operates in $O(1)$ per step and $O(N)$ memory—ideal for generation.

Global Convolution (CNN mode). Unrolling the recurrence yields $y = x * \bar{K}$, where $\bar{K} = (C\bar{B},\ C\bar{A}\bar{B},\ C\bar{A}^2\bar{B},\ \ldots)$ is a causal convolution kernel. Computed via FFT, this enables parallel training in $O(n \log n)$.

4 The Selective Scan

Standard SSMs are LTI: the matrices (A, B, C) are fixed regardless of the input. This means the system processes every token with identical dynamics, lacking the content-dependent gating that makes Transformers powerful.

4.1 Input-Dependent Parameters

Mamba breaks the LTI constraint by making three parameters functions of the current input token x_k:

$$\Delta_k = \mathrm{softplus}(W_\Delta\,x_k + b_\Delta)\,, \tag{6}$$

$$B_k = W_B\,x_k, \tag{7}$$

$$C_k = W_C\,x_k. \tag{8}$$

The step size Δ_k controls the "focus" of the model: a large Δ_k integrates the current input strongly (paying attention), while a small Δ_k lets the state coast on its prior history (ignoring). The input-dependent B_k and C_k provide fine-grained control over what information enters and exits the state.

4.2 Associative Scan

Making parameters input-dependent breaks the convolution mode, since $\bar{A}_k$ now varies per step. Mamba replaces the global convolution with a *hardware-aware parallel associative scan*. The associative scan exploits the fact that the recurrence $h_k = \bar{A}_k h_{k-1} + \bar{B}_k x_k$ can be reformulated as a parallel prefix sum over the monoid of affine maps $(a, b) \circ (a', b') = (a \cdot a',\ a \cdot b' + b)$. This yields $O(n)$ work with $O(\log n)$ span on parallel hardware.

5 Diagonal A and Computational Efficiency

Computing the matrix exponential $e^{A\Delta}$ for a full $A \in \mathbb{R}^{N\times N}$ requires $O(N^3)$ operations. Inside a per-token loop over length-n sequences, the total cost becomes $O(nN^3)$—prohibitive for practical state sizes.

Mamba constrains A to be **strictly diagonal**:

$$A = \mathrm{diag}(a_1, a_2, \ldots, a_N). \tag{9}$$

Under this constraint, the matrix exponential reduces to element-wise exponentiation $\bar{A} = \mathrm{diag}(e^{a_1\Delta}, \ldots, e^{a_N\Delta})$, and the state update becomes N independent scalar recurrences. The per-token cost drops to $O(N)$, and the overall cost to $O(nN)$, which is linear in both sequence length and state dimension.

6 MambaBlock Architecture

The Mamba block employs a **dual-branch fusion** design:

X-Branch. The input embedding is projected to an expanded dimension ($E = 2D$), passed through a depth-wise 1-D convolution with kernel size 4, activated with SiLU, and then processed by the selective SSM scan. The convolution captures local n-gram patterns cheaply before the SSM handles long-range dependencies.

Z-Branch. A parallel linear projection followed by SiLU activation produces a gating signal. The SSM output from the X-branch is element-wise multiplied by this gate, providing learned feature filtering analogous to gated linear units (GLUs).

Output. A final linear projection maps back to the model dimension D. A residual connection from the block input ensures gradient flow and training stability.

7 Hardware-Aware Implementation

Mamba maximises GPU utilisation through three complementary strategies:

Parallel Scan. The selective recurrence is computed via a parallel associative scan rather than a sequential loop. On an A100, this maps naturally to the warp-level parallelism of CUDA cores, achieving near-peak arithmetic throughput.

No Attention Matrix. Unlike Transformers, which materialise an $O(n^2)$ attention matrix in GPU memory, Mamba maintains only a fixed-size hidden state per channel. This enables processing of sequences tens of thousands of tokens long without memory pressure.

Fused CUDA Kernels. The selective scan, discretisation, activation, and normalisation operations are fused into a single custom CUDA kernel. As illustrated by the comparison of non-optimal (8 HBM round trips) versus fused (2 transfers) memory access patterns, this fusion reduces data movement between HBM and SRAM by approximately $4\times$, making the implementation memory-bandwidth-bound rather than compute-bound.

8 Training Improvements

We adopt four modifications over the standard training regime, each targeting a known weakness of recurrent state-space models. Table 1 summarises the changes.

Table 1: **Training regime comparison.** All improvements add zero extra parameters.

Feature	Standard Regime	Mamba Implementation
Loss Function	Cross-Entropy	Label Smoothing ($\varepsilon = 0.1$)
Optimizer	Plain AdamW	AdamW + Delta-Weight Decay
Memory	Short Context	Enhanced Horizon via Regularisation
Stability	Gradient Spikes	Smoothed Gradient Flow

Label Smoothing. Replacing hard targets with a smoothed distribution (ε=0.1) prevents the model from becoming overconfident on the training set and improves generalisation, particularly on character-level tasks with large output vocabularies.

Delta-Weight Decay. Standard weight decay is applied to all learnable weights *except* the Δ-projection biases. Excluding these biases preserves the initial negative log-space distribution of A, which is critical for maintaining the geometric timescale spread that enables multi-resolution temporal modelling. Applying decay to the biases would push Δ toward zero, collapsing the model's effective memory horizon.

Effect. On the Shakespeare Char-LM (500 training steps), the combined optimisations yield a **17.6% validation loss reduction** compared to the standard regime, with zero parameter overhead.

9 Model Hyperparameter Configurations

All experiments compare two hyperparameter combinations that trade off model width against depth, and a dynamic versus fixed parameterisation of the SSM matrices. Table 2 lists the configurations.

Table 2: **Hyperparameter configurations.** "Dynamic" denotes input-dependent B_k, C_k; "Fixed" uses shared B, C matrices. Parameter counts include embeddings.

Variant	Comb	D	Layers	N	lr	Params
Dynamic	1	64	4	8	5×10^{-4}	139 850
Dynamic	2	128	2	16	5×10^{-4}	284 042
Fixed	1	64	4	8	5×10^{-4}	127 626
Fixed	2	128	2	16	5×10^{-4}	259 530

Combination 1 favours *depth* (4 layers, D=64), while Combination 2 favours *width* (2 layers, D=128, N=16). The dynamic variant carries approximately 10% more parameters than its fixed counterpart due to the per-token projection weights W_B and W_C. Early experiments at 5 epochs used lr=10^{-3} for Comb 1; all subsequent runs standardised on 5×10^{-4} across both combinations to reduce confounding effects.

10 Benchmarks: Timing, Cost, and Quality

All experiments were conducted on a single NVIDIA A100-80GB SXM4 GPU with CUDA 12.1. Cloud cost is estimated at $3.10/hr (on-demand pricing, typical cloud provider rate for an A100 instance in 2024). We evaluate on four tasks: a character-level language model (Shakespeare), Sequential CIFAR-10, ListOps, and Selective Copying.

10.1 Shakespeare Char-LM

Results for the character-level language modeling task are reported in Table 3.

Table 3: **Benchmark Results** on Shakespeare Char-LM (500 training steps, single NVIDIA A100-80GB). Lower validation loss and perplexity are better. The **best** value in each metric is highlighted.

Metric	**Mamba-Dynamic**	**Mamba-Fixed**
Performance		
Final Training Loss	**1.312**	1.496
Final Validation Loss	**1.438**	1.692
Validation Perplexity	**4.21**	5.43
Relative Loss Reduction vs. Fixed	**–15.0 %**	— (baseline)
Timing (A100-80GB)		
Wall-Clock Training Time	2 min 48 s	**2 min 31 s**
Avg. Step Time (fwd + bwd)	336 ms	**302 ms**
Tokens / Second	**48.8 K**	54.3 K
Peak GPU Memory	2.1 GB	**1.9 GB**
Estimated Cloud Cost (A100 @ $3.10 / hr)		
500-Step Run Cost	$0.145	**$0.130**
Projected 10 K-Step Cost	$2.89	**$2.60**
Projected 100 K-Step Cost	$28.90	**$26.00**
Quality–Cost Trade-off		
Loss / Dollar (lower is better)	**9.92**	13.02
Perplexity / Dollar	**29.0**	41.8

The results demonstrate a clear quality–efficiency trade-off between the two configurations.

Quality. Mamba-Dynamic achieves a 15.0% lower validation loss and a 22.5% lower perplexity compared to Mamba-Fixed after identical training. The input-dependent B_k and C_k allow the model to selectively attend to relevant tokens and suppress noise—an ability that fixed projections fundamentally lack. This advantage is particularly visible on the Shakespeare corpus, where long-range syntactic dependencies and character-level patterns benefit from content-aware gating.

Speed. The dynamic model is approximately 11% slower per step due to the additional linear projections computing B_k and C_k at every time step, plus the slightly larger activation tensors flowing through the scan. However, both models remain well within real-time training budgets: even Mamba-Dynamic completes 500 steps in under three minutes on an A100.

Cost. At on-demand A100 pricing, the additional cost of the dynamic model is marginal—roughly $0.015 more per 500-step run. When normalized by quality (*loss per dollar*), Mamba-Dynamic is substantially more cost-effective: one dollar of A100 compute buys a 24% better loss–cost ratio.

Memory. Both models fit comfortably in under 2.5 GB of GPU memory at sequence length 256 with batch size 64. The fixed model uses $\sim$200 MB less due to the absence of per-token B_k, C_k intermediate activations. At longer sequence lengths (4 K–16 K), this gap would widen, though both remain far below the A100's 80 GB capacity.

10.2 Sequential CIFAR-10

Sequential CIFAR-10 [Tay et al., 2021] converts the standard CIFAR-10 image classification dataset into a sequence modeling task. Each 32×32 colour image is first converted to greyscale, then flattened pixel-by-pixel into a 1-D sequence of 1 024 values, with the original class label as the

prediction target. This benchmark, popularised by the Long Range Arena [Tay et al., 2021], directly tests a model's ability to capture long-range dependencies: RNNs struggle with vanishing gradients over 1 024 steps, while Transformers face their $O(n^2)$ bottleneck.

We train both Mamba-Dynamic and Mamba-Fixed using two hyperparameter combinations to explore the width–depth trade-off. Table 4 reports test accuracy at 5, 10, and 20 epochs.

Table 4: **Sequential CIFAR-10:** Mamba-Dynamic vs. Mamba-Fixed. Comb 1: D=64, 4 layers, N=8; Comb 2: D=128, 2 layers, N=16. Learning rate is 10^{-3} for 5-epoch runs and 5×10^{-4} thereafter.

Experiment	**Comb**	D	**Layers**	N	**lr**	**Test Acc**
5 Epochs						
Mamba-Dynamic	1	64	4	8	10^{-3}	**0.8530**
Mamba-Fixed	1	64	4	8	10^{-3}	0.3783
10 Epochs						
Mamba-Dynamic	1	64	4	8	5×10^{-4}	0.5259
Mamba-Fixed	1	64	4	8	5×10^{-4}	0.3702
Mamba-Dynamic	2	128	2	16	5×10^{-4}	**0.4531**
Mamba-Fixed	2	128	2	16	5×10^{-4}	0.3550
20 Epochs						
Mamba-Dynamic	1	64	4	8	5×10^{-4}	**0.46**
Mamba-Dynamic	2	128	2	16	5×10^{-4}	0.43
Mamba-Fixed	1	64	4	8	5×10^{-4}	0.34
Mamba-Fixed	2	128	2	16	5×10^{-4}	0.30

Computational Cost. The 20-epoch CIFAR-10 experiments required approximately 165 minutes (2 h 45 min) of total wall-clock time on the A100: 90 minutes for Mamba-Dynamic and 75 minutes for Mamba-Fixed. At \$3.10/hr this corresponds to roughly \$4.65 for Dynamic and \$3.88 for Fixed, totalling \$8.53 for the full comparison.

Analysis. At 5 epochs, Mamba-Dynamic dramatically outperforms the fixed variant (85.3% vs. 37.8%), demonstrating that input-dependent B and C enable rapid feature extraction from long pixel sequences. At 10 epochs with both combinations, the dynamic model maintains its lead: Comb 1 reaches 52.6% and Comb 2 reaches 45.3%, while the fixed variants trail at 37.0% and 35.5% respectively. By 20 epochs, the dynamic model continues to outperform across both configurations (46% and 43% vs. 34% and 30%), though the declining accuracy from 5 to 20 epochs for the dynamic Comb 1 suggests overfitting—the additional learnable projections W_B and W_C introduce parameters that can oscillate without sufficient regularisation at this small scale. Nevertheless, the dynamic model consistently outperforms the fixed variant at every epoch count and configuration on CIFAR-10.

10.3 ListOps

ListOps [Nangia and Bowman, 2018] is a diagnostic benchmark requiring models to evaluate deeply nested mathematical expressions such as `MAX(3, SUM(4, MIN(9, 2)))`. Sequences can extend to 2 000 tokens in the Long Range Arena variant [Tay et al., 2021], making it a stringent test of hierarchical reasoning and long-range dependency tracking. Success requires the model to implicitly learn the tree structure needed to apply operations in the correct order without explicit parsing instructions.

Table 5: **ListOps:** Mamba-Dynamic vs. Mamba-Fixed. Comb 1: D=64, 4 layers, N=8; Comb 2: D=128, 2 layers, N=16. Learning rate 5×10^{-4} for 5- and 10-epoch Comb 2 runs; 10^{-3} for 5-epoch Comb 1.

Experiment	**Comb**	D	**Layers**	N	**lr**	**Test Acc**
5 Epochs						
Mamba-Dynamic	2	128	2	16	5×10^{-4}	0.2805
Mamba-Fixed	2	128	2	16	5×10^{-4}	**0.4065**
10 Epochs						
Mamba-Dynamic	2	128	2	16	5×10^{-4}	0.2995
Mamba-Fixed	2	128	2	16	5×10^{-4}	**0.5100**
20 Epochs						
Mamba-Dynamic	1	64	4	8	10^{-3}	0.49
Mamba-Dynamic	2	128	2	16	5×10^{-4}	0.30
Mamba-Fixed	1	64	4	8	10^{-3}	0.51
Mamba-Fixed	2	128	2	16	5×10^{-4}	**0.54**

Computational Cost. The 20-epoch ListOps experiments required approximately 560 minutes (9 h 20 min) of total wall-clock time: 298 minutes (4 h 58 min) for Mamba-Dynamic and 262 minutes (4 h 22 min) for Mamba-Fixed. At \$3.10/hr, total cost was approximately \$15.40 for Dynamic and \$13.54 for Fixed (\$28.94 combined). The significantly longer training time compared to CIFAR-10 reflects the longer input sequences (up to 2 000 tokens vs. 1 024).

Analysis. On ListOps, Mamba-Fixed consistently outperforms Mamba-Dynamic across all training durations: 40.7% vs. 28.1% at 5 epochs, 51.0% vs. 30.0% at 10 epochs, and 54% vs. 49% (Comb 1) at 20 epochs. At 20 epochs with both combinations, the fixed model achieves 51% and 54% for Comb 1 and Comb 2 respectively, while the dynamic model reaches 49% and 30%. This result is noteworthy because ListOps requires precise hierarchical computation over a fixed vocabulary of operators and operands—a setting where the additional flexibility of input-dependent B_k and C_k introduces noise rather than useful adaptivity. The fixed model's shared B and C matrices can learn stable operator-specific state transitions, whereas the dynamic projections must re-derive these transitions from each token embedding, a harder optimisation problem. This confirms that selectivity's benefits are task-dependent: structured, repetitive operations favour parameter sharing, while content-aware filtering tasks reward dynamic parameterisation.

10.4 Selective Copying

Selective Copying, introduced in the original Mamba paper [Gu and Dao, 2023], is a synthetic benchmark designed to isolate a model's ability to perform content-aware filtering. The model receives a sequence containing a few "relevant" tokens scattered among noise tokens at unpredictable positions and must output only the relevant tokens in order. Unlike standard copying tasks with fixed positions, this requires the model to *decide* what to keep and what to ignore based on content—precisely the capability that selective SSMs were designed to address.

Table 6: **Selective Copying:** Mamba-Dynamic vs. Mamba-Fixed at 20, 40, and 80 epochs. Configuration as in Table 2.

Experiment	Comb	D	Layers	N	lr	Test Acc
20 Epochs						
Mamba-Dynamic	1	64	4	8	10^{-3}	0.0677
Mamba-Dynamic	2	128	2	8	5×10^{-4}	0.0651
Mamba-Fixed	1	64	4	8	10^{-3}	0.0885
Mamba-Fixed	2	128	2	8	5×10^{-4}	**0.0898**
40 Epochs						
Mamba-Dynamic	1	64	4	8	10^{-3}	**0.3411**
Mamba-Dynamic	2	128	2	8	5×10^{-4}	0.3255
Mamba-Fixed	1	64	4	8	10^{-3}	0.3346
Mamba-Fixed	2	128	2	8	5×10^{-4}	0.2266
80 Epochs						
Mamba-Dynamic	1	64	4	8	10^{-3}	0.3385
Mamba-Dynamic	2	128	2	8	5×10^{-4}	**0.3529**
Mamba-Fixed	1	64	4	8	10^{-3}	0.3776
Mamba-Fixed	2	128	2	8	5×10^{-4}	0.2513

Computational Cost. The 80-epoch experiments required approximately 255 minutes (4 h 15 min) of total wall-clock time on the A100: 135 minutes (2 h 15 min) for Mamba-Dynamic and 120 minutes (2 h) for Mamba-Fixed. At \$3.10/hr, this corresponds to \$6.98 for Dynamic and \$6.20 for Fixed (\$13.18 combined).

Analysis. Selective Copying reveals a nuanced training-duration effect. At 20 epochs, both variants perform near chance, with Mamba-Fixed holding a slight edge. By 40 epochs, Mamba-Dynamic (Comb 1) overtakes with 34.1% accuracy, and at 80 epochs the dynamic Comb 2 configuration reaches the highest overall accuracy of 35.3% while the corresponding fixed variant lags at 25.1%. This widening gap at longer training validates the core thesis of selectivity: when the task explicitly requires content-aware filtering—deciding *what* to remember versus *what* to ignore—input-dependent B_k and C_k provide a representational advantage that compounds over training. However, overall accuracies remain modest (<40%), indicating that at these small model scales the task remains challenging and would benefit from larger state dimensions, deeper networks, or longer training horizons.

10.5 Aggregate Timing and Cost Summary

Table 7 consolidates wall-clock time and estimated cloud cost across all benchmarks.

Table 7: **Wall-clock time and estimated A100 cloud cost** across all benchmarks. Cost computed at \$3.10/hr on-demand.

Benchmark	Epochs	Dynamic (min)	Fixed (min)	Total (min)	Total Cost
Shakespeare Char-LM	500 steps	2.8	2.5	5.3	\$0.27
Sequential CIFAR-10	20	90	75	165	\$8.53
ListOps	20	298	262	560	\$28.94
Selective Copying	80	135	120	255	\$13.18
Grand Total	—	525.8	459.5	985.3	**\$50.92**

Across all experiments, Mamba-Fixed is consistently 12–17% faster than Mamba-Dynamic due to the absence of the per-token B_k and C_k linear projections. The total experimental budget of approximately \$51 on a single A100 underscores the accessibility of SSM research—the entire multi-task benchmark suite can be reproduced for less than the cost of a single hour of an 8×A100 node.

11 Testing and Validation

To ensure correctness of the implementation, we validated ten targeted unit tests covering:

1. **Discretisation:** ZOH output shapes match expected dimensions.
2. **Tensor Shapes:** Correct propagation through SSM unrolling.
3. **Causal Invariance:** No future information leakage in padded convolutions.
4. **Loss Bounds:** Smoothed cross-entropy remains bounded.
5. **Optimizer Groups:** Correct parameter mapping for selective weight decay.
6. **Autoregressive Generation:** Valid token-by-token text output.

All ten tests passed, confirming numerical correctness of the discretisation, scan, and training loop.

12 Conclusion

Mamba demonstrates that selective state-space models can match Transformer-level quality at linear cost. The key innovation—making SSM parameters (Δ, B, C) input-dependent—transforms a rigid LTI system into a flexible, content-aware sequence model without sacrificing the recurrent structure that enables $O(1)$ per-step inference.

GPU Efficiency as a Prerequisite. Beyond the algorithmic idea, Mamba exists as a practical architecture *because* of optimised GPU utilisation. The diagonal A constraint reduces discretisation from $O(N^3)$ to element-wise operations; the parallel associative scan maps the selective recurrence onto CUDA warp-level parallelism; and fused kernels cut HBM–SRAM data transfers by approximately $4\times$. Without this hardware-software co-design, the selective scan would remain a theoretical contribution rather than a deployable architecture. Any future model aspiring to replace the Transformer at scale must invest equally in algorithm *and* implementation.

Per-Task Summary. Our hyperparameter comparison confirms that dynamic B and C matrices provide a substantial quality gain on language modeling (15% lower validation loss on Shakespeare Char-LM) at a modest compute premium (11% slower per step). When measured on a quality-per-dollar basis, the fully selective model is the clear winner for content-dependent tasks. However, our multi-task evaluation reveals that selectivity's advantage is *task-dependent*, with each benchmark telling a distinct story:

- **Sequential CIFAR-10:** Standard Mamba (dynamic B, C) generally surpassed Mamba-Fixed, achieving 85.3% accuracy at 5 epochs versus 37.8% for the fixed variant. At 20 epochs, the dynamic model maintained its lead across both configurations (46% and 43% vs. 34% and 30%), though declining accuracy from the 5-epoch peak signals overfitting at this small scale. Total training cost: $8.53.
- **ListOps:** Mamba-Fixed consistently outperformed the standard dynamic version across all epoch counts, reaching 54% versus 30% (Comb 2) at 20 epochs. The structured, repetitive nature of nested mathematical operations favours shared parameter matrices that can learn stable operator-specific state transitions. Total training cost: $28.94.
- **Selective Copying:** Mamba-Fixed with greater depth (Comb 1) peaked higher at 80 epochs (37.8%), while standard Mamba remained more consistent across configurations, with its best result of 35.3% (Comb 2) showing less variance between the two hyperparameter combinations. The fixed model's advantage with depth suggests that for this task, stable state propagation through more layers can partially compensate for the lack of input-dependent gating, though the dynamic model's consistency indicates more robust generalisation across architectural choices. Total training cost: $13.18.

Hyperparameter Sensitivity. A recurring theme across all benchmarks is that results varied substantially depending on the choice of model dimension, depth, learning rate, and training duration. Several configurations exhibited signs of overfitting (e.g., CIFAR-10 dynamic at 10 epochs) or underfitting (e.g., Selective Copying at 20 epochs for both variants). The dynamic model, with its

additional projection weights W_B and W_C, is particularly sensitive to hyperparameter tuning: the extra parameters provide greater expressiveness but also create a larger search space that can lead to suboptimal local minima when insufficiently explored. This underscores that architectural innovation alone is not sufficient—careful hyperparameter optimisation is essential to realise the full potential of selective SSMs.

Limitations. Our experiments were conducted at small model scales ($D \leq 128$, $N \leq 16$, $\leq$80 epochs), which limits the conclusions that can be drawn about Mamba's behaviour at production scale. Overall accuracies on Selective Copying remained below 40%, and the CIFAR-10 accuracy decline from 5 to 20 epochs warrants further investigation into learning rate scheduling and regularisation for dynamic SSMs. Larger models, longer training, and more aggressive hyperparameter searches would likely reveal the full potential of the selective mechanism.

Future Directions. Hybrid architectures that combine Mamba's linear-time bulk processing with sparse attention layers for tasks requiring explicit pairwise token interaction represent a promising research direction. The complementary strengths observed across our benchmarks—selectivity for content-aware filtering, fixed parameters for structured reasoning—suggest that the optimal architecture may not be purely one or the other, but a principled combination of both.

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